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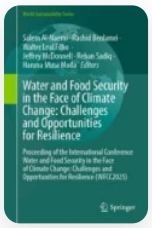

Water and Food Security in the Face of Climate Change: Challenges and Opportunities for Resilience

(WFCC 2025)

Riadh Bechir & Mohamed Jaoued

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Abstract

The Province of Médenine and thereafter Oum Zessar watershed are characterized by the aridity of the climate, and the fragility of the plant communities. The agrarian system itself is characterised by the coexistence of irrigated and dry agriculture (cereal, fruit ...) with pastoralism (rangeland). Conversely, land uses have changed with the privatization of the traditional collective lands and the evolution towards a more intensive agro-pastoral system. Due to the increased human needs and the development of agriculture, the pressure on natural resources, and particularly on land resources, is becoming higher and higher. This pressure induces land degradation and a significant decrease of crop yield. The sedentarisation of pastoralism and an accelerated land privatization, as well as governmental subsidies for some new agricultural activities like irrigated agriculture, have caused land fragmentation and increasing pressure on the land. This pressure is driven by demographic growth and an increase in human consumption, but also by new investments and some speculations from external actors. In this context, the traditional agrarian system which was managing the access to natural resources was substituted by new competing systems for natural resources, in particular, water resources. This paper reports on a participatory impact assessment of alternative soil and water conservation (SWC) scenarios in the Oum Zessar watershed, Tunisia. The first objective was to assess the impact of three SWC scenarios on key social, economic and environmental land use functions and climate change adaptation. The second objective was to test and evaluate the applicability of the 'Framework for Participatory Impact Assessment for assessing scenario impacts in the context of a developing country, in this case Tunisia. Framework for Participatory Impact Assessment is an innovative tool for carrying out a sustainability impact assessment of CES scenarios and for studying sustainable development issues. We conclude that participatory impact assessment contributes to a better regional understanding of key links between policy effects and sustainable development, which forms the basis for better policy decision-making.

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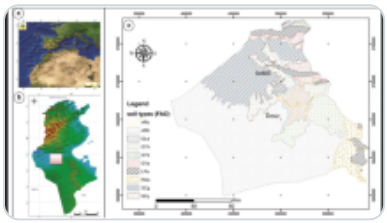

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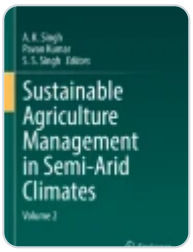

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